

Yoruba vs English Deictic Deep Dive

This handout distills the linguistic analysis behind the response differences in the Yoruba deixis experiment. The key claim is that Yoruba encodes deictic anchoring more explicitly than English through aspect, existential location, adverbial urgency stacking, and overt subject positioning, and that these features interact with prompting constraints to shape response type.

FOCUS

Deictic markers

LANGUAGES

Yoruba / English

CORE ISSUE

Response type shift

MAIN CLAIM

Grammar × prompt

Central Claim

Yoruba supplies a grammatically denser deictic field than English. Through markers like *ń*, *máa ń*, *wà*, and stacked temporal adverbials, it makes the origo more explicit, more proximal, and more embodied. The instruction layer then determines whether the model answers inside that field with compressed verdicts or expands outward into expository ethics discourse.

Marker Families

Aspectual Markers

ti, ń, and máa ń do not simply map onto English tense-aspect choices. They often add event vividness, present relevance, or narrative proximity.

- ti: can strengthen completed-from-now evaluation pressure.
- ń: foregrounds ongoingness and can make the event feel live.
- máa ń: makes repeated behavior feel actively recurrent in the present frame.

Locative / Existential Markers

wà, níbí, níbè, and láti ipò yìí localize the speaker or scene more overtly than English often needs to.

- Reflexive Yoruba often literally places the self in the scene.
- Spatial Yoruba builds a stronger scene map than English here/there contrasts.
- Temporal Yoruba stacks urgency adverbials more densely than English.

Why Trolley Is So Important

The trolley impersonal prompt is classified in the prompt JSON as double_progressive_irreducible because the opening uses ń sáré ń sún mó. This is not a minor translation detail. It stages the event as unfolding and proximate, making detached classroom explanation harder to sustain and making action pressure easier to trigger.

Prompt-Type Consequences

Yoruba feature	English contrast	Likely response effect
ti completive	English can stay narratively flatter	More already-evaluated event frame; easier move toward recommendation

Yoruba feature	English contrast	Likely response effect
ń progressive	English urgency can stay looser	Higher proximal pressure; stronger sense of unfolding moral emergency
máa ń habitual	English habitual can feel more neutral	Repeated live behavior; more agent-recognition and moral immediacy
wà existential-locative	English need not localize the subject	More embodied self-implication and conscience rhetoric
Temporal adverb stacking	English can use a leaner timing frame	Urgency saturation; less room for extended hesitation
Overt pronoun anchoring	English can remain more expository	Stronger address, responsibility, or collectivity effects

What The Data Show

Constrained Yoruba

Short, verdict-heavy, highly compressed outputs. Deictic pressure stays visible, but the format layer forces it into decision-first rhetoric.

- More direct recommendations
- Less framework listing
- Higher interventionist skew in trolley

Unrestricted Yoruba

Longer, more expository outputs. The same Yoruba markers remain, but the models are free to expand into ethical pedagogy, translation behavior, or mixed discourse.

- More framework narration
- More hesitation and scene explanation
- More drift in weaker models, especially DeepSeek

Most Important Methodological Point

The experiment should not be described as if “Yoruba responses” form a single homogeneous type. The impersonal condition alone splits into multiple subtypes in the machine-readable prompt typology:

- bare_perfective
- progressive_background
- double_progressive_irreducible
- bare_perfective_with_habitual
- existential_stative_restructured

This means some Yoruba “impersonal” prompts are structurally closer to English than others. The analysis should therefore group by subtype, not treat all six impersonal cells as interchangeable.

Bottom Line

It is wrong to say that the differences are only caused by Yoruba grammar, and it is also wrong to say they are only caused by prompting. The strongest conclusion is an interactional one: Yoruba grammatical resources define a denser and more embodied deictic terrain than English, and the response instruction layer decides whether the model speaks inside that terrain as a compressed decision-maker or as an expansive ethics commentator.

Source note: this handout is derived from `outputs/yoruba_english_deictic_deep_dive_20260529.md`.